

SETI and Policy

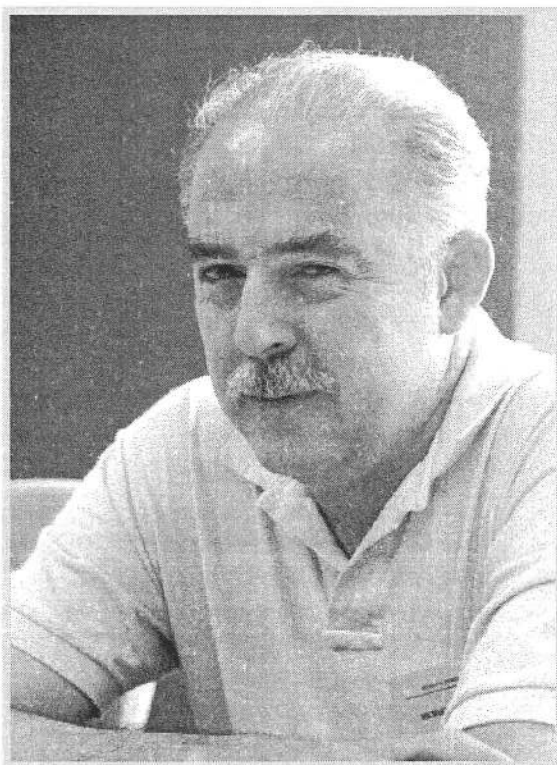
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We have seen in the previous chapter that different groups may respond in different ways to news of a detection depending upon circumstances as well as group characteristics. Governments will have to respond to a detection and to subsequent events, and some groups will seek to influence the governmental decision-making processes. This section outlines some ways in which political groups and governments might react, including how governments could act at the international level. While such reactions are not entirely predictable, we suggest here some policy approaches that could be adopted at national and international levels to manage political responses in considered and effective ways.

Political Reactions to an ETI Signal Detection

Domestic and National Reactions

The distances and times involved in interstellar communication are such that there is no essential immediacy in contemplating a reply, should humanity detect an ETI signal or message. However, there may well be a measure of urgency within political systems in deciding how governing bodies might react or publicly respond to the detection. There is no exact analog in our experience that would help



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us predict political reactions to this event.

The SETI-related literature mentions studies of political reactions to events such as nuclear accidents and earthquakes (Logsdon and Anderson, 1990), but these events are qualitatively different from an ETI detection.

The political reactions to detection of a signal would depend on the circumstances that pre-

vail at the time and also on the nature of the signal itself. At one extreme, the signal may be little more than the equivalent of a dial tone, containing no information beyond the fact that its senders exist. As an intermediate possibility a signal may contain information, but the information may be very limited, difficult to decipher, or ambiguous when deciphered. This would be an "information-poor" signal. At the other extreme a signal might be "information-rich", comprehensible and delivered in a form that gives us a lot of unambiguous information. The nature and form of an ETI transmission, wherever it might lie within this spectrum of possibilities, will influence political reactions to the signal and will determine the timing, nature, and content of any contemplated reply that may be sent from Earth.

All of the various social groups discussed in Chapter 3 could compete for political recognition and power to serve their goals in one way or another. Many could be expected to react in ways intended to capitalize on (or to discredit) the new factor. Religious groups or scientists, for example, might also see the detection as an opportunity to enhance their own political influence. Political parties and political leaders would certainly want to capitalize on the event and its attendant publicity for political gain (or prevent others from capitalizing on it). Governments could attempt to control or monopolize the information in efforts to maximize their political benefit. Significant partisan political consequences could result from successful or unsuccessful management of the information that accompanies detection. If the signal is information-rich, firms, consortia, and other economic entities might compete for access to ETI information and might try to limit the access of competitors to that information. Some parts of national economies, perhaps including aerospace and defense industries, might see the

detection of a signal as an argument for continued or enhanced commitment of resources to their programs.

Some political entities in the nation making the discovery might disagree with the view that information about the detection should be shared with other nations. They may well argue that the discovering nation – particularly if it had made a substantial investment in the search – should have prior access (if not exclusive rights) to the information. Thus, information about an ETI detection is a potential stimulus for many political manipulations, especially if the signal is information-rich and takes a long time to decipher.

Efforts to limit early information dissemination are not likely to be successful, given the extensive scientific network involved and the need for the first observers of a genuine signal to obtain verification by their scientific colleagues. The 1958 act that created NASA and the Freedom of Information Act also created a legal expectation in the United States that information acquired by NASA programs will be made available to the public. Despite legal safeguards, the assurance by NASA that all information will be released, and knowledge of the practical impossibility of limiting public access to information, it is possible that some groups, individuals, or institutions are likely to claim for their own purposes that information is being withheld. (Note that, if the detection were made by a private organization rather than by a governmental program, the information might not be in the public domain.)

If the detected signal contained a message of some kind, the political impact would be greater and would persist much longer than if it were information-poor. The impacts would depend on the content of the message and the time needed to decipher it. If the message could not be quickly interpreted, accusations

that information had been intentionally withheld or manipulated would be likely, potentially fueling fears that the message contained threats to Earth that were being kept from the public. The ubiquity of science fiction films, many of which present extraterrestrial intelligence as threatening, would, for a large segment of the population, lend a certain plausibility to this view.

Various groups could use suspicion of a cover-up for their own purposes. As discussed in Chapter 3, millennial or even paranoid responses are possible by certain groups, some of which might try to provoke public anxieties and capitalize on them. The possibility of such responses is sufficiently high to justify preparation and planning for responses designed to forestall adverse public reactions akin to that following the Orson Welles radio broadcast of a Martian invasion. Scientists and the international scientific community could be particularly important in such a situation by explaining details of the received signal and reasons for the length of time needed for deciphering it, and by maintaining a sense of objectivity and calm professional "presence" in a situation ripe for exploitation and distortion.

The issue of when to release information, even with an essential commitment to openness, raises difficult questions that could cause conflict with the established, cautious practices of the scientific community. (This is discussed in detail in Chapter 5.) Despite the approach employed and no matter how much care is exercised, there is no way to prevent accusations from skeptical groups (or those who choose to use the issue for their own purposes) that information is being manipulated.

International Reactions

Immediate international political effects would surround the drama of the initial announcement of detection of intelligent life elsewhere in the universe. The announcement by itself could translate into a measure of international prestige comparable to that enjoyed by the United States when the Voyager spacecraft sent images from the outer planets or when the Viking spacecraft reported results of life detection experiments on Mars. Prestige would accrue not only to the nation making the first detection and announcement but also to that country in which the primary detection technology was located. The host country would be likely to have a considerable interest in sharing the resulting international attention. This initial political prestige would probably not persist for long if the signal had no information content and if the civilization from which it came were many tens or hundreds of light years away. As a restraining factor, the intense public attention attracted by a detection announcement would result in considerable embarrassment if the claim of detection later proved to be unfounded.

In most foreseeable scenarios there is likely to be substantial ambiguity surrounding a possible detection. A great deal of dialog and debate within the scientific community would probably be required before any widely-accepted conclusion could be reached that a signal was artificial and of ETI origin (see Chapter 5). If so, the element of surprise would be much reduced and there would be less immediate political value to any of the countries involved, when the long-delayed conclusion was finally announced.

In the special case of detection of an information-rich signal that is weak and not immediately decipherable, a concerted (and possibly

costly) effort might be required to improve the signal-to-noise ratio and to conduct a potentially time-consuming program of decoding. In such circumstances, the existence of the signal is likely to receive continued public attention and the nation taking the lead in the decoding effort could capitalize on the political benefits of that attention.

It is possible, though unlikely, that the detection of a very weak signal (or a signal in a region of the spectrum not easily observed by most instruments) could lead to a situation in which only one nation had the technical means to study the signal. While such a unique capability could eventually be duplicated by other nations, a detecting nation that excluded international participation might enjoy an advantage for some time, possibly provoking resentment or political maneuvering by other nations.

The unequal ability of nations to participate in SETI or in the excitement and activities surrounding a detection might be a further irritant in existing political divisions, especially in North-South relations. It is also possible that a government or a non-governmental institution controlling a powerful transmitter might seek to jam or interfere with the signal for various political or ideological reasons. While technically difficult outside a local area, such an action would be likely to call for political (and possibly other) responses by other affected governments.

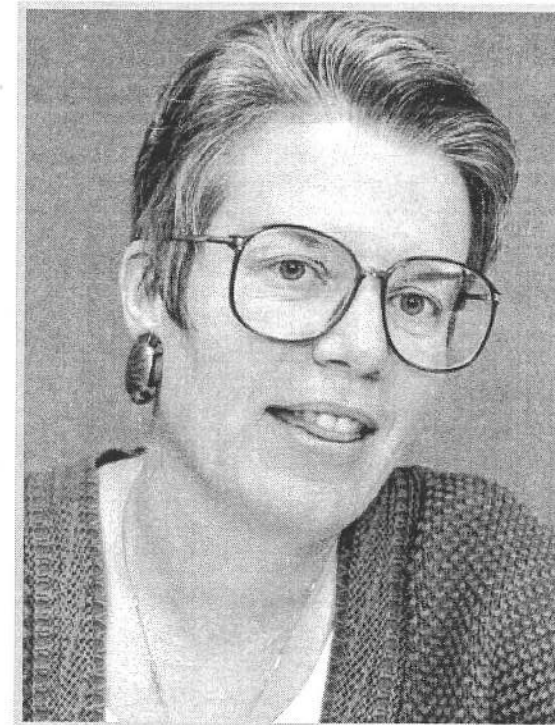
More optimistically, the receipt of an ETI signal and its consequent decipherment might contribute to the emergence of a "global village" atmosphere. Here a sense of humanity's common identity, as contrasted with the extra-terrestrial identity of the senders of the signal, could serve to put racial and ethnic differences on Earth in a new perspective. This effect might persist if political leadership emerged

that was able to capitalize on the new mood and move disparate and contentious political forces toward greater international cooperation, conciliation, and resolution of differences.

In the past, perceived threats or opportunities that appear to involve most of humankind (such as proliferation of nuclear weapons or destruction of the ozone layer as a result of human activities) have had limited but significant effects on international political cooperation. Nations have built substantial institutions and mechanisms for international cooperation in the following instances:

- To counter certain threats (for example, nuclear weapons proliferation or smallpox);
- When cooperation was necessary to exploit a technology (for example, the world-wide allocation of the radio frequency spectrum to competing services);
- When evident externalities had to be dealt with (for example, in limiting ocean dumping);
- When costs were too high for one nation to bear (for example, particle physics in Europe or the development of a magnetic fusion reactor);
- When exploration or exploitation of resources beyond national territories was contemplated (for example, the Antarctic and Outer Space treaties).

In the absence of an imminent threat (or prospect of immediate gain for humanity) from a detection of ETI, there is little reason to expect that any new sense of shared human destiny that might follow detection



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would last long enough to create substantial prospects for new levels of cooperation. It is more likely that there would be little lasting political change in the near term.

While it is not possible to identify with any certainty the political or social groups that would gain in power, prestige, or influence as a result of an ETI detection, we may speculate that some groups might try to exploit such a situation for their advantage. These include the governments of countries directly involved, those of other countries with advanced space and communications capabilities, scientists and industries engaged in astronomical and space research programs, advocates of UFO's, practitioners of channeling and other mystical beliefs involving alien civilizations, and perhaps millenarian and apoca-

lyptic groups in general. Whether the claims of these groups would be effective would depend upon the details of the detection and on presently-unpredictable responses by governments, media, the public, and other social groups. It does not seem possible to identify specific groups that would lose power or prestige other than by comparison with those that would gain.

Long Term Implications

In the very long run, the new realization that life on Earth is not unique in the universe would undoubtedly have cultural (and consequently political) effects, perhaps of comparable power to those generated by the earlier revolutions due to Copernicus and Darwin. This would seem particularly likely if the signal received were from a civilization close enough that a two-way communication could be begun, even if question-and-answer exchanges would necessarily require many years.

It might be asked what political effects might result from a search program that continues for years without detecting any signal that can be confidently attributed to ETI's. Obviously that result would not prove that there is no other intelligent life in the universe. Over time, however, consistently negative results of SETI programs might reinforce a conviction that intelligent life on the Earth is in fact unique. Perhaps that would eventually lead to greater realization of the significance of human life and the importance of preserving what would then appear to be the sole example of intelligent life in the universe.

Existing Processes for Handling a Detection

At present we know of no formally institutionalized processes at national or international levels for addressing a discovery of extraterrestrial intelligence.

In 1977, the question of extraterrestrial civilizations was raised in the United Nations Committee on the Peaceful Uses of Outer Space. A number of delegations addressed the question, including those of Austria, Czechoslovakia, and Indonesia, with the last proposing that the question be assigned to the Scientific and Technical Subcommittee (United Nations document A/AC.105/206, 1977). In 1978, the General Assembly briefly considered the question and invited interested countries to coordinate their scientific research on extraterrestrial life and to inform the Secretary General of the results of their activities (General Assembly decision 33/426, 1978). However, there was little response to this invitation and the subject has not been raised since then.

At the international level, international law and practice do not address SETI issues directly. However existing agreements may provide some broad guidelines that could be extrapolated to address the formulation of an initial international response to an ETI detection. Major international agreements that form the basis for international space law already contain some principles and norms that could be interpreted to provide limited guidance. Perhaps the most relevant is Article XI of the 1967 Outer Space Treaty, which states that:

States parties to the treaty will inform the Secretary General of the United Nations as well as the public and the international sci-

entific community, to the greatest extent feasible and practicable, of the nature, conduct, locations and results of their activities in outer space relating to its peaceful use (United Nations, 1967).

This article also provides that the Secretary General should be prepared to disseminate such information immediately and effectively.

The 1979 Agreement Governing Activities of States on the Moon and Other Celestial Bodies (to which the United States and most other industrialized nations are not parties, and to which only eight nations adhered during the first 13 years of its existence) states that:

In carrying out activities under this agreement, States parties shall promptly inform the Secretary-General, as well as the public and the international scientific community, of any phenomena they discover in outer space, including the Moon, which could endanger human life or health, as well as of any indication of organic life (United Nations, 1979).

This is as close as current international law gets to addressing the question of extraterrestrial life forms.

In the international regulatory field, the Radio Regulations of the International Telecommunication Union (ITU) reserve some frequencies for communication to and from spacecraft and for radio astronomy and space research. The Regulations include a footnote stating that research is being conducted by some countries that search for intentional transmissions of extraterrestrial origin (ITU, 1982).

While such guidance may be helpful in a broad sense, the fact remains that governments have not agreed how they should deal with contact with an extraterrestrial civilization.

Although it is not an intergovernmental agreement, the *Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence* (hereafter the *Declaration*) takes a small step in that direction. The *Declaration*, endorsed by several international non-governmental scientific and space organizations, grew out of a series of discussions within the SETI Committee of the International Academy of Astronautics and the International Institute of Space Law. This document is open for signature by individuals, agencies, or organizations that conduct scientific searches for extraterrestrial intelligence. Signatories voluntarily agree to observe certain principles if they believe they may have detected ETI's, above all to:

- (1) Verify in cooperation with other signatories that what they have found is credible evidence of ETI, and
- (2) Announce any verified discovery and make information about it available to the public.

In addition, signatories agree to observe other specified scientific procedures in their treatment of data. The *Declaration* states that no messages should be sent in reply to any detected ETI signals without appropriate international consultations. The procedures for such consultations are not set out in the *Declaration* but would be the subject of another declaration or agreement.

The *Declaration* is not legally binding and has no enforcement provisions. However, there is an incentive for SETI organizations and researchers to sign, in that signatories get early access to information about any discovery. The text of the *Declaration* is given in Appendix A.

Issues Governments Would Face

Given the absence of applicable laws, procedures and policies, governments supporting SETI programs should develop plans for actions they would take if a detection of ETI occurred. Outlined below are some procedural and institutional steps that might be taken at the national level to better prepare for and manage the aftermath of a detection of extraterrestrial intelligence. To illustrate the issues involved, we use the United States of America as an example. We recognize that other nations may have different procedures and institutional considerations, though we expect that many of them will be similar in nature.

Any governmental organization conducting SETI searches, including NASA in the US, needs to establish a post-detection verification and notification process. Within the US government, NASA would want to notify the Executive Office of the President, the Department of State, NASA's Authorization and Appropriations committees in both the Senate and the House of Representatives, the Foreign Affairs and the Intelligence committees of both Houses, and the Congressional leaders of the two major political parties. As a way of beginning, it would also be useful for NASA to endorse (by signing) the *Declaration*, as it delineates some global standards for post-detection activities.

In the event that a signal is detected, policy for regular dissemination of information about the signal to the public would be necessary. This requires identifying an office or institution responsible for providing data to scientists and the public. The National Space Science Data Center at NASA's Goddard Spaceflight Center might be a logical location

at which to archive the data and make them available to users. One possible US model for data archiving and sharing is the policy announced in 1991 for data on global climate change (National Science Foundation, 1991).

Government decision-makers in general should be informed about SETI research and its implications so that they are better prepared to anticipate and deal with subsequent events if there is a signal detection. Means of doing this would include briefings for selected officials and legislators and the broad distribution of publications on SETI and its implications.

A list of technical experts who would be available to provide briefings to decision-makers and the media on short notice could be useful. NASA would be a logical agency to assemble and maintain such a list.

Further analyses of the consequences of contact, including the political, economic, and sociological dimensions of possible public and institutional responses, would be helpful. Possible sponsoring institutions include NASA and the National Science Foundation.

In the best of worlds, activities following detection of an ETI signal would largely involve activation of processes set up in anticipation of detection. The level of effort needed to establish those procedures is not high, and a modest investment in preparation now would help to ensure a calm and measured public and political response to a detection in the future.

Issues at the International Level

At the international level, governments and international organizations would also benefit from advance preparations for the event. These preparations could include several measures. Agencies involved in SETI research should inform officials of appropriate international organizations, and national representatives to those organizations, about SETI and its possible outcomes. Briefings, publications, and discussions would be appropriate to convey the information. In the case of the United Nations, a logical approach would be to have NASA brief the Scientific and Technical Subcommittee (STS) of the Committee on the Peaceful Uses of Outer Space (COPUOS). The International Astronautical Federation and the Committee on Space Research would also make presentations to the STS, as these non-governmental organizations have expertise and can offer independent advice. The STS would then be in a position to report on its discussion to the full COPUOS and the General Assembly. The International Telecommunication Union (ITU), which regulates the use of the radio frequency spectrum, should also be informed in advance.

Agencies involved in SETI research, together with their sponsoring national governments, could consult with agencies and officials of international organizations to identify a channel for formal notification of appropriate international organizations in the event of a confirmed detection. In the case of notification to the United Nations, this would require that a member state notify the Secretary General through its foreign ministry or its representatives to the UN. Similar channels might be identified for notifying other international bodies, such as the International

Telecommunication Union, as quick action to protect certain radio frequencies might be required. The agencies might also formally or informally agree in advance upon the level of ETI signal confirmation required to initiate official intergovernmental notification and the content of such notification. (The latter might include information on the source of the detected ETI signal, the nature of the signal, and contacts for further information.)

The United Nations has routine procedures for disseminating official notifications submitted to the Secretary-General by member governments. These take the form of an official document to all governments. (The process is slow. We note that the extraordinary event of an ETI detection, which would be well reported in the press, might require more rapid handling.) Appropriate international scientific bodies (such as the Committee on Space Research) have already been identified as organizations to be informed in the Declaration.

The government of the discovering nation could commence discussions of possible follow-up actions within appropriate international organizations. In the case of the United Nations, this might require a special session of COPUOS (which is not in session most of the year) with later discussion in the General Assembly. Special studies and reports (whose nature would be dictated by the circumstances of the detection and any information derivable from the signal) might be commissioned and the results submitted to the United Nations for distribution to members of COPUOS for consideration at the next meeting. Appropriate bodies of the ITU could also recommend actions (such as protecting the frequency of the extraterrestrial signal from interference).

Agencies currently involved in SETI research could make efforts to broaden international



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participation in research and the exchange of information concerning SETI. These agencies could provide training and technical assistance to interested scientists of nations that are potentially able to participate in SETI research. Guest scientists from countries without SETI programs could be invited to participate in SETI research in host countries in order to improve their ability to discuss SETI issues with their respective publics, to broaden informed dialog, and to contribute to the research. Agencies of nations with the most advanced SETI capabilities, such as NASA in the case of the United States, might take the lead in these endeavors, possibly in cooperation with appropriate national or international organizations.

The United Nations, with its specialized agencies, is not the only intergovernmental organization that might have to address the conse-

quences of a detection. In some areas, regional organizations play a major policy role. The most obvious example is Western Europe, where the institutions of the European Community might become involved in determining the way the region responds to a detection.

The Question of a Reply, and the Conduct of Relations

If we detect a signal from an ETI civilization, we would need to consider whether to try to contact that civilization (if the signal is unintentional) or to send a message in reply (if the signal is a message directed to humanity). This question would of course be more urgent if the detected signal contained decipherable information and/or the other civilization were relatively nearby (for example, a few tens rather than thousands of light-years away).

The prospect of sending a reply raises many questions. Are there reasons not to reply? Who decides whether or not we should do so? Who would be responsible for formulating a reply? What should be the content of a reply? Should individual nation states (or institutions within them) act independently in making these decisions or should humanity act through one international organization or process?

One or more member states could raise the issue in COPUOS and propose that the Committee recommend a certain policy or course of action to the General Assembly. If the proposal is accepted by the Committee through its consensus procedures and then adopted by the General Assembly in the form of a resolution, the proposal would become international policy. As a General Assembly resolution rather than a formal treaty or legal

agreement, however, the policy would not be legally binding. The formal negotiations and national legislative ratifications required for a legally binding inter-governmental agreement are difficult and time-consuming and are not a priority for most governments. Efforts towards a legally binding agreement would therefore not appear feasible for the foreseeable future.

If an international agreement on procedures for determining policy on communications with extraterrestrial civilizations exists at the time of detection, the government of the discovering nation would initiate those procedures upon announcing the discovery. Depending on the details of that agreement and the discovery, international (or other) discussions could address the question of whether or not to send a message to the other civilization. If a decision were taken to send a message, further discussions would ultimately define its timing and content and identify an optimal mode of transmission. Proposals could simultaneously be made to the ITU concerning use of the relevant frequency channels and their protection by international agreement.

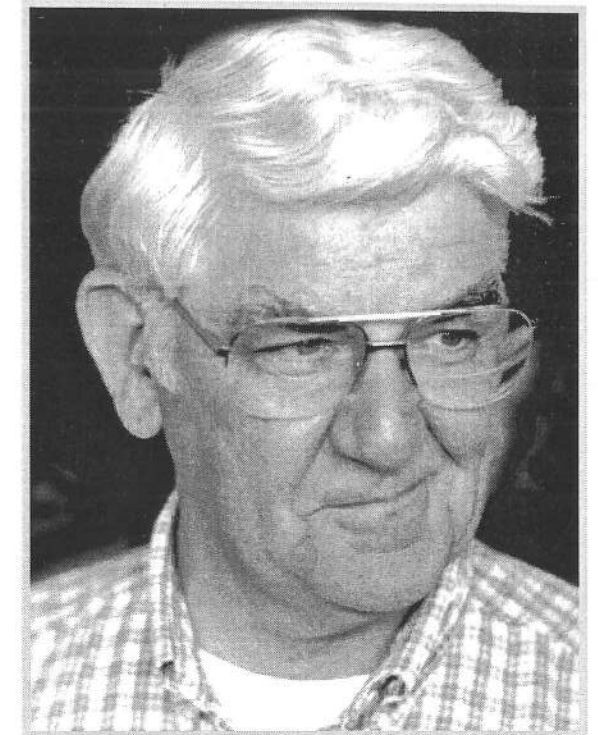
It will be difficult to sustain interest in the subject of a reply at the intergovernmental level in the long absence of a detection. However, a few particularly germane international non-governmental organizations might begin the process by discussing proposals with regard to a reply. The results of those discussions could then be communicated to governments and other appropriate international organizations for their own consideration or to quickly "bring them up to speed" in the event of a detection.

Entering into a communicating relationship with another civilization could raise questions about the long-term conduct of relations. We

have developed many mutually acceptable practices for the conduct of relations among our national states. Over the centuries we have acquired considerable knowledge of how to do this in the human context. We have also developed legal concepts and practices that codify many aspects of these relationships. Entering into communication with ETI's would require that humankind develop many new approaches to diplomacy and law. Who is in charge of the conduct of such relations - individual nation states or an international institution? Who will be responsible for formulating and sending future messages, after the first reply? What should be the contents of future messages? Responses to these questions may evolve from preliminary discussion of an agreement on communication with extraterrestrial intelligence, but a more permanent institutional approach may emerge from the actual experience of a detection.

More philosophically, how do we conduct relations with intelligent beings sprung from different evolutions, existing in different environments, with different languages and thought processes, possessing values and interests that may be very different from our own? At a minimum we will need political and cultural sensitivity and tolerance for broad differences in perspective and physical form. To sustain a communicating relationship, we will also need a broader perspective on our own species and a significant degree of consensus on our own interests and purposes. Consensus within the human community would be particularly necessary if we agreed to formulate and transmit single unified messages from humankind to the detected ETI's.

There is little published work on the question of relations between different intelligent species and that work is inevitably speculative. Further thought might prove fruitful. In addition, broad studies of the evolution of



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diplomacy among humans from this fundamental perspective might provide useful insights.

A few scholars and lawyers have speculated about principles for the conduct of such a relationship. Legal scholar Andrew Haley described a concept of relations he called "Metalaw" based on a revised version of the Golden Rule (Haley, 1963). Robinson and White (1979) extended Haley's concept in 1985 and suggested a concept based on a presumption of moral and legal equality among all intelligent life forms.

These beginnings make it clear that much thought and preparation would be needed if we are to eventually engage in the conduct of



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long-term relations with ETI. At the same time, it could be a mistake to formalize approaches to communication and the conduct of relations into law and treaties before persuasive evidence of extraterrestrial intelligence is actually found. Czech legal scholar Vladimir Kopal, former Director of the Outer Space Affairs Division of the United Nations, has written that

It would not be wise to request an immediate consideration of a special regulation governing the SETI/CETI activities... The consideration of a legislative process can only be started when the boundary between possibilities and well-established realities has been crossed (Kopal, 1990).

Recommendations

Because existing institutions, processes, and agreements generally neglect the possibility of an ETI detection, it is clear that much remains to be done to prepare for policy responses to a detection. Work could be addressed in two broad categories: first begin the educational and informational processes necessary to prepare government bodies and international institutions most likely to be affected by an ETI detection, and second begin establishing procedures and mechanisms similar to those outlined in this chapter to make humankind's responses to detection more effective. Specifically, we recommend:

- (1) Establish post-detection notification processes within and among concerned national governments and inter-governmental bodies;
- (2) Establish policy and procedures on the public dissemination of information about the detection by agencies sponsoring searches, governments, and international organizations;
- (3) Inform national governments and international organizations about SETI and the implications of detection;
- (4) Conduct further analyses of the political consequences of detection;
- (5) Work with governments and international organizations to develop national and international procedures for the responses to detection;
- (6) Broaden international participation in SETI through training and technical assistance.